CUSTOMER & PROCUREMENT INTELLIGENCE

NPO Elektromashina (НПО «Электромашина»)

Reconstruction of a Russian Industrial Customer Network

Russian Federation

Subject	NPO Elektromashina (НПО «Электромашина»)
Jurisdiction	Russian Federation
Case type	Customer / Procurement Intelligence
Source basis	Public Russian-language sources

RESEARCH QUESTION

Identify and verify the commercial/customer network of a Russian industrial manufacturer using procurement records, technical documentation, corporate disclosures and independent sources. The research objective was not to produce a conventional corporate due-diligence profile. Instead, the investigation focused on reconstructing the manufacturer's customer and procurement footprint, identifying the products supplied where possible, distinguishing direct buyers from related/group entities and downstream users, and assessing the strength of the available evidence.

Prepared from public sources only. No private, paid or non-public data was used.

Research Question Identify and verify the commercial/customer network of a Russian industrial manufacturer using procurement records, technical documentation, corporate disclosures and independent sources. The research objective was not to produce a conventional corporate due-diligence profile. Instead, the investigation focused on reconstructing the manufacturer's customer and procurement footprint , identifying the products supplied where possible, distinguishing direct buyers from related/group entities and downstream users, and assessing the strength of the available evidence.

1. Executive Summary

This investigation reconstructed the publicly visible customer network of AO «Научно-производственное объединение «Электромашина» (NPO Elektromashina) , a Chelyabinsk industrial manufacturer with a strong specialization in electrical equipment and documented activity in defense, industrial and civilian transport applications. Public procurement aggregators based on EIS data identify 35 customer organizations and 208 supplier-side contracts involving NPO Elektromashina, with an aggregate recorded value of approximately RUB 1.592 billion . The five largest customer relationships by recorded contract value are NPK Uralvagonzavod, SKB Turbina, Omsktransmash, 61 BTRZ and 81 BTRZ. The investigation was able to establish several relationships at a substantially higher evidentiary level than a simple customer-list approach. In particular, public records directly connect Elektromashina with:

- NPK Uralvagonzavod for electrospecial equipment;
- 61 BTRZ for the DEA-10T diesel power unit and APP-172 transmission selector;
- 81 BTRZ for components for the BTR-82AM;
- SKB Turbina for D418N electromagnets;
- Russian Ministry of Defence for specific electrospecial-equipment items including BU25-2S, VKU-1 and R15M-3S;
- Kurganmashzavod for BU088-1S / 6EC088-1S and 960-M DMB-4-M;
- Omsktransmash for the AMASIS industrial lighting project;
- Uraltransmash for the EDBT1 traction motor used on the 71-415R retro-tram;
- PK Transport Systems for TAED72 traction motors used in Vityaz-M trams. The resulting network is not a homogeneous set of independent commercial customers. A substantial part of the identified footprint is embedded in the Uralvagonzavod / Rostec industrial ecosystem , while another part consists of government entities and civilian

industrial relationships. Historical corporate disclosures show that NPO Elektromashina became a subsidiary of the Uralvagonzavod group in 2015, with the group's ownership reported at 91.61% at 30 June 2015 and 31 December 2015. A key finding is that transaction role and corporate affiliation cannot be treated as the same variable . For example, Omsktransmash and Uraltransmash can be direct buyers while simultaneously being group companies. Conversely, a downstream military unit can be an end user without being the direct procurement counterparty. The Kurganmashzavod litigation provides a clear example of this distinction: Elektromashina supplied Kurganmashzavod, while a military unit was identified as the downstream end user of the supplied equipment. Overall, the evidence supports a high-confidence reconstruction of a multi-layer industrial customer network , but not a complete reconstruction of every historical transaction or every end-use application. Public procurement records frequently use generic descriptions such as "supply of products" or "supply of items", while technical specifications are not always publicly indexed. Accordingly, the final network deliberately distinguishes verified product relationships from unresolved product attribution .

2. Target and Research Scope

Target Entity

АО «Научно-производственное объединение «Электромашина» NPO Elektromashina

- OGRN: 1047422507736
- INN: 7449044990
- Registered: 27 July 2004
- Registered address: 454119, Chelyabinsk, ul. Mashinostroiteley 2. Corporate disclosure and current Rostec material identify the company as part of the Uralvagonzavod concern within Rostec . Historical Uralvagonzavod reporting also shows that Elektromashina entered the group as a subsidiary during the 2015 ownership

restructuring. Research period Primary analytical period: 2016-29 September 2026 Historical evidence predating 2016 is retained where it materially helps establish the development of a customer relationship or corporate structure. Customer definition The research distinguishes: Direct customer The organization directly contracts with NPO Elektromashina as buyer. Group company A direct customer that belongs to the same corporate industrial group or corporate contour. Related company A legally related entity that should not be treated as an independent external customer. Intermediary / distributor A counterparty purchasing for onward sale, integration or distribution. No such relationship was conclusively established in the present evidence set. End user The downstream organization actually operating the equipment. Unverified A claimed or database-level relationship for which sufficient transactional evidence has not yet been established.

3. Initial Customer Universe

The procurement footprint currently visible in public commercial/procurement aggregators contains 35 customer organizations and 208 supplier-side contracts with an aggregate recorded value of approximately RUB 1.592 billion . The largest customer relationships by recorded contract value are: Customer Contracts Recorded value NPK Uralvagonzavod 9 RUB 455.99m SKB Turbina 57 RUB 304.59m Omsktransmash 9 RUB 293.24m 61 BTRZ 4 RUB 172.83m 81 BTRZ 10 RUB 64.16m VNIITRANSMASH 1 RUB 55.48m CNII Burevestnik 23 RUB 46.57m 560 BTRZ 2 RUB 26.00m 144 BTRZ 10 RUB 25.98m UBT-Ecology 3 RUB 22.39m MChS Chelyabinsk Region 4 RUB 18.16m MK Vityaz 13 RUB 17.65m 103 Arsenal 12 RUB 16.90m Russian Ministry of Defence 2 RUB 15.61m CNIITochmash 2 RUB 10.63m Source: procurement data derived from EIS and aggregated by Synapse. This list should be treated as an initial customer universe , not as a final list of 35 independent commercial customers. Corporate affiliation, historical ownership and the distinction between internal and external relationships must be resolved before the network is interpreted analytically.

4. Verified Customer/Product Relationships

4.1 NPK Uralvagonzavod

A public EIS-derived procurement record identifies contract 5662302953815000891000, concluded on 8 April 2015, for the supply of electrospecial-equipment sets to NPK Uralvagonzavod. Note: the group-company status recorded elsewhere in this report is dated 30 June 2015, so as at the contract date it is not established.Note: the group-company status recorded elsewhere in this report is dated 30 June 2015, so as at the contract date it is not established.The recorded value is approximately RUB 286.88 million . A later contract, 56623029538180004450000 , dated 27 April 2018, records a further supply of electrical products worth approximately RUB 119.80 million . Established chain NPO Elektromashina → electrospecial-equipment sets → NPK Uralvagonzavod Limitation The public record does not expose enough information to identify the exact vehicle, platform or final application associated with the 2015 contract. Therefore: Product category: High confidence. Specific end application: Not established. It would be methodologically incorrect to infer a specific armored-vehicle application merely from Uralvagonzavod's industrial profile.

4.2 61 BTRZ

Contract 77819310752150000550000 , dated 6 July 2015 , explicitly identifies two products:

- DEA-10T diesel power unit
- APP-172 automatic transmission selector The recorded contract value is approximately RUB 165.20 million . Independent technical material also identifies APP-172 as a product manufactured by Elektromashina and describes its use in armored-vehicle applications. This independently supports the product/manufacturer attribution, but does not by itself prove the exact platform

for the 2015 61 BTRZ transaction. Established chain NPO Elektromashina → DEA-10T + APP-172 → 61 BTRZ Confidence High for transaction/product identity. Medium for specific end platform of this particular transaction.

4.3 81 BTRZ

Contract 72302060955160001800000 , dated 8 July 2016 , explicitly identifies: components for BTR-82AM The recorded value is approximately RUB 51.99 million . Established chain NPO Elektromashina → components → 81 BTRZ → BTR-82AM Limitation The public contract description does not identify the exact component-level nomenclature. Therefore the evidence supports the platform/application, but not a specific component. Confidence: High.

4.4 SKB Turbina

Contract 6745203381515000289000 0 , dated 27 December 2015 , explicitly concerns the: manufacture and supply of D418N electromagnets with a recorded value of approximately RUB 1.09 million . This is a direct product-level procurement record. A separate arbitration case independently confirms that Elektromashina and SKB Turbina had a direct supplier-buyer relationship and records another supply relationship involving ASEP12 . The court evidence therefore independently corroborates the commercial relationship , but it should not be used as evidence that the court case itself concerned D418N. This distinction matters. Established chain NPO Elektromashina → D418N electromagnet → SKB Turbina Defensive qualification The D418N transaction is directly supported by procurement evidence. The judicial record independently confirms a supplier-buyer relationship between the same entities, but concerns a different product. Confidence: High for D418N transaction.

5. Russian Ministry of Defence

The procurement record contains multiple highly specific defense-related transactions. Contract 1770425226117000666 , dated 7 July 2017 , concerns supply of engines, spare parts, assemblies and armored-equipment property, explicitly including: blocks and electrospecial-equipment boxes for armored vehicles. Recorded value: approximately RUB 13.66 million . An earlier contract, 0173100004514001049 , dated 25 August 2014 , concerns overhaul of electrospecial equipment for armored vehicles, explicitly identifying BU 25-2S ; the recorded value is approximately RUB 1.95 million . Patent documentation independently identifies R15M-3S as a voltage regulator manufactured by Elektromashina and describes its electrical function. Established chain NPO Elektromashina → BU25-2S / VKU-1 / R15M-3S and related electrospecial equipment → Ministry of Defence → armored-vehicle applications Evidence assessment The product identity is strongly supported. The procurement relationship is strongly supported. The exact downstream vehicle model for every individual item is not established in the available public record . Confidence: High for product and defense-use category.

6. Kurganmashzavod

Kurganmashzavod is one of the most valuable relationships in the investigation because it can be corroborated through court records rather than procurement databases alone . An Arbitration Court decision establishes that on 31 May 2017 , Kurganmashzavod and Elektromashina entered into contract: 171918741282241224101540 3 / 150D-0171-17/00 The court confirms actual deliveries under the contract through delivery notes dated in 2018 and 2019. The litigation specifically identifies a defective product as BU088-1S , part of 6EC088-1S . This is substantially stronger than a simple procurement-table entry because the independent judicial source documents the actual contractual relationship and delivery history. Established chain NPO Elektromashina → BU088-1S / 6EC088-1S → Kurganmashzavod Confidence: High.

6.1 Downstream end user: Military Unit 41450

A separate Arbitration Court record establishes that Elektromashina supplied 960-M DMB-4-M to Kurganmashzavod. During operation, a defect involving voltage regulator RN10-1S was identified at the downstream end-user site: Military Unit 41450, Ryazan The court explicitly describes the military unit as the location of the end user's equipment and records replacement of the faulty regulator. Established chain NPO Elektromashina → 960-M DMB-4-M → Kurganmashzavod → Military Unit 41450 This is an important example of why: Direct buyer ≠ end user. The military unit should not be counted as an independent direct customer of Elektromashina based on this evidence.

7. Omsktransmash

The procurement database identifies three substantial contracts:

- 6550520417117000039 0000
- 10 February 2016 - approximately RUB 92.02m
- 6550520417117000186 0000
- 3 October 2016 - approximately RUB 95.66m
- 6550520417117000187 0000
- 10 January 2017 - approximately RUB 48.25m The procurement descriptions are generic and simply refer to "supply of products." However, an independent manufacturer/project disclosure provides much stronger product-level information. In February 2021, Elektromashina implemented a modernization project at Omsktransmash

involving LED industrial lighting equipped with the AMASIS intelligent-control system . Elektromashina states that it performed working-design development, equipment supply, electrical installation and commissioning. Omsktransmash's chief power engineer is quoted describing the project's operation. An additional AMASIS page independently publishes an Omsktransmash testimonial confirming

successful implementation in production shops. Established chain NPO Elektromashina → AMASIS intelligent lighting → Omsktransmash → industrial production-shop lighting Confidence: High. Important qualification The evidence does not establish that AMASIS was the subject of the three large 2016-2017 procurement contracts. The 2021 project should therefore be treated as a separate independently verified customer relationship , not retrospectively attached to an earlier generic procurement record.

8. Uraltransmash

An independent industry source states that the 71-415R retro-tram , produced by Uraltransmash, uses the EDBT1 electric motor manufactured by NPO Elektromashina . The source further describes technical characteristics and the vehicle's testing. Rostec independently reports the development and use of Elektromashina's tram propulsion technology in the Uraltransmash ecosystem. Established chain NPO Elektromashina → EDBT1 traction motor → Uraltransmash → 71-415R retro-tram Limitation The product/customer relationship is independently established, but the evidence found does not directly tie this exact motor installation to one of the procurement-contract identifiers in the customer database. Confidence: Medium-High.

9. PK Transport Systems

This is the strongest contemporary civilian relationship in the dataset. Rostec reports that in April 2021 Elektromashina supplied an initial batch of six TAED72 traction motors for the Vityaz-M tram manufactured by PK Transport Systems. The motors were subsequently tested in tram operation. The 2021 disclosure describes the technical task and testing process. A July 2026 Rostec publication reports that:

- further motors were supplied in 2023;
- the motors completed approximately 90,000 km in testing;
- PK Transport Systems ordered a modified version for southern regions;
- in July 2026 the companies signed a new cooperation agreement for further supply of traction asynchronous motors. Established chain NPO Elektromashina → TAED72 traction motor → PK Transport Systems → Vityaz-M tram Confidence High. This relationship also demonstrates that Elektromashina's publicly visible customer network

is not limited to defense-industrial demand; it includes an active civilian electric-transport segment.

10. Relationship-Type Resolution

The customer network should not be interpreted as 35 independent external customers. Corporate relationships and transaction roles overlap. Group-company relationships Historical Uralvagonzavod reporting identifies NPO Elektromashina as a major subsidiary and separately identifies Omsktransmash, Uraltransmash, UKBTM, CNII Burevestnik and Zavod No. 9 among Uralvagonzavod's major subsidiaries. Accordingly, several customer relationships should be classified as: Group company / direct customer rather than simply "independent customer." Examples include:

- NPK Uralvagonzavod
- Omsktransmash
- Uraltransmash
- UKBTM
- CNII Burevestnik
- Zavod No. 9 Other Rostec companies Other identified customers belong to different Rostec structures and should not be merged into the UVZ subgroup merely because they share the ultimate state-owned parent.
- Examples include:
- SKB Turbina

- KBP
- VNIИ Signal
- VOMZ
- NPO GIPO
- NPP Start
- NPO Poisk
- CNIITochmash
- The proper network representation is therefore: NPO Elektromashina → Direct customer → Other Rostec entity
- rather than: NPO Elektromashina → UVZ subsidiary unless the specific corporate relationship is independently established. Related company OOO TD Elektromashina is a special case. The procurement database records a supply contract 77449130449170000140000 , dated 18 October 2017, concerning electrical equipment for Russian Railways. However, the entity is

identified as a 100%-controlled subsidiary of NPO Elektromashina and was subsequently liquidated. This means it should not be counted as an independent external customer. Correct classification: Related company / internal trading channel Intermediaries / distributors No relationship has been conclusively classified as an intermediary or distributor on the present evidence. End users The strongest established downstream end-user case is: Kurganmashzavod → Military Unit 41450 The end user should remain a separate graph node rather than being counted among direct customers.

11. Final Evidence Matrix

Only relationships that can be supported by evidence are included below. Customer Product Evidence Date Value Relation ship type Confid ence NPK Uralvagonza vod Electrosp ecial-equi pment sets EIS procurement contract 5662302953815000 8910000 08.04.2 015 RUB 286.8 8m Group company / direct customer High NPK Uralvagonza vod Electrical products, exact nomenclat ure not exposed EIS contract 5662302953818000 4450000 27.04.2 018 RUB 119.8 0m Group company / direct customer High 61 BTRZ DEA-10T + APP-172 EIS contract 7781931075215000 0550000 06.07.2 015 RUB 165.2 0m Group company / direct customer High 81 BTRZ Compone nts for BTR-82A M EIS contract 7230206095516000 1800000 08.07.2 016 RUB 51.99 m Group company / direct customer High SKB Turbina D418N electroma gnets EIS contract 6745203381515000 2890000 27.12.2 015 RUB 1.09m Direct customer / other Rostec High Ministry of Defence BU25-2S and other electrospe cial equipment EIS contracts 0173100004514001 049 and related records; patent independently confirms R15M-3S manufacture 2014-2 017 Multip le contra cts Governm ent direct customer High Kurganmash zavod BU088-1S / 6EC088-1 S Arbitration court confirms contract and actual deliveries 31.05.2 017 + deliverie s 2018-2 019 Not disclo sed Group company / direct customer High Kurganmash zavod 960-M DMB-4-M Arbitration court confirms supply Historic al contract relations hip; court decision 2023 Not disclo sed Group company / direct customer High Military Unit 41450 Downstre am use of 960-M DMB-4-M Arbitration record identifies end-user location 2020 defect record - End user High Omsktransm ash AMASIS industrial lighting system Manufacturer/projec t disclosure + Omsktransmash testimonial 2021 Not disclo sed Group company / direct customer High Uraltransma sh EDBT1 traction motor Independent industry source identifies EDBT1 as Elektromashina product installed in 71-415R 2021 Not disclo sed Group company / direct customer Medium -High PK Transport Systems TAED72 traction motors Rostec corporate disclosures; 2021 supply, later deliveries and 2026 agreement 2021-2 026 Not disclo sed Direct customer High

12. Analytical Findings

Finding 1 - The visible customer network is strongly embedded in the Russian state-industrial ecosystem

The procurement universe contains numerous organizations associated with the Uralvagonzavod/Rostec and broader defense-industrial structure. The largest customer by recorded value is NPK Uralvagonzavod, while Omsktransmash, Uraltransmash and CNII Burevestnik are documented Uralvagonzavod subsidiaries in historical corporate reporting. This means the procurement footprint should not be interpreted as a conventional SME-style diversified customer portfolio consisting primarily of unrelated commercial buyers. Assessment High confidence.

Finding 2 - The product portfolio connects the manufacturer to multiple layers of the industrial value chain

The evidence covers at least three distinct product/application domains: Defense and armored vehicles

- DEA-10T
- APP-172
- BTR-82AM components
- BU25-2S
- VKU-1
- R15M-3S
- BU088-1S / 6EC088-1S
- 960-M DMB-4-M Industrial infrastructure
- AMASIS intelligent lighting
- LED industrial lighting and control systems Civilian electric transport
- EDBT1
- TAED72 This indicates that the publicly visible customer footprint is not merely a reflection of one defense product family. The evidence shows a broader industrial engineering portfolio with a continuing civilian-transport segment. Assessment High confidence.

Finding 3 - Procurement data alone materially

under-describe product relationships

Several of the largest procurement relationships are represented in public databases by descriptions such as: "supply of products" or "supply of items." This is particularly visible in the Omsktransmash and VNIITRANSMASH records. Product attribution therefore required cross-referencing procurement records with corporate disclosures, judicial documentation, technical material and product information. This is a methodological result of the investigation, not merely a limitation of the dataset. Assessment High confidence.

Finding 4 - Court records substantially improve

customer intelligence quality

The Kurganmashzavod cases show the analytical value of litigation data. A procurement database establishes the existence of a supplier relationship. A court record can additionally establish:

- the contractual parties;
- contract number;
- contract date;
- actual delivery;
- specific product;
- defect/reclamation history;
- and, in one case, the downstream end user. This makes judicial records particularly valuable for reconstructing relationships that procurement aggregators describe only superficially.

13. Hypotheses

The following statements are analytical hypotheses , not direct factual findings.

Hypothesis 1 - A significant portion of

Elektromashina's historical customer base is

structurally dependent on the wider UVZ/Rostec

industrial ecosystem.

Basis Multiple major customers are documented as UVZ subsidiaries or other Rostec entities, while NPO Elektromashina itself is a UVZ enterprise. Confidence High. The available data strongly support the existence of a concentrated state-industrial ecosystem around the company. What remains unresolved is the economic proportion of external versus intra-group revenue , because the public customer data do not provide a sufficiently clean consolidated revenue segmentation.

Hypothesis 2 - Elektromashina operates not only as a component supplier but also as a systems/engineering integrator in selected civilian applications.

Basis The AMASIS project at Omsktransmash involved not merely product delivery but working-design development, equipment supply, installation, commissioning and personnel training. Confidence High for the specific AMASIS project. The broader claim that such turnkey integration represents a major share of the company's business remains unverified .

Hypothesis 3 - Procurement records materially understate the complexity of the true customer network.

Basis Some records identify only a generic "product" or "item," while other sources reveal a detailed product, downstream platform or end user. Confidence High. The conclusion concerns the quality and granularity of public data , not an assumption about undisclosed transactions.

Hypothesis 4 - Some apparently separate customers may represent internal industrial supply chains rather than independent market demand.

Basis The customer universe includes multiple companies within the same industrial group, and at least one recorded "customer" - TD Elektromashina - is a controlled internal trading entity rather than an independent buyer. Confidence High for the existence of internal-chain relationships. The scale of such internal demand across the entire network remains unquantified .

14. Evidence Gaps and Unresolved Questions

Several gaps remain and should be preserved in the final case rather than artificially closed. Product attribution gaps For a number of high-value customer relationships, especially Uralvagonzavod and VNIITRANSMASH, public procurement records confirm a commercial transaction but do not expose the complete product nomenclature. End-user gaps Even where an immediate buyer is identified, the final end user is often not visible. The Kurganmashzavod case demonstrates that downstream users can exist without appearing as the direct procurement counterparty. Contract-chain gaps A product can be independently associated with a customer without finding the precise contract through which that product was supplied. The Uraltransmash / EDBT1 relationship is an example: the product and platform connection are strongly documented, but the exact procurement identifier was not established in the evidence set used here. Corporate-affiliation gaps Corporate group structures change over time. A company classified as a UVZ subsidiary during one reporting period should not automatically be assigned the same status for every year of the research period. Historical ownership therefore needs to be evaluated as-of date , not as a timeless attribute. Aggregator limitations The 35-customer universe is highly useful for discovery, but the aggregator should not be treated as the underlying primary source. Where possible, the final evidentiary hierarchy should prioritize

- 1. EIS / procurement record;
- 2. court record;
- 3. customer disclosure;
- 4. manufacturer disclosure;
- 5. government / industry source;
- 6. secondary aggregator.

15. Final Analytical Conclusion

The investigation establishes that NPO Elektromashina has a multi-layered publicly observable customer network extending across defense-industrial enterprises, government procurement and civilian industrial transport . The strongest evidence supports direct commercial relationships with organizations including NPK Uralvagonzavod, 61 BTRZ, 81 BTRZ, SKB Turbina, Kurganmashzavod, Omsktransmash, Uraltransmash, the Russian Ministry of Defence and PK Transport Systems . Several of these relationships can be linked to identifiable products and applications rather than merely to a company name. The resulting network has three analytically distinct characteristics. First, it is strongly embedded in the Russian state-industrial and Rostec ecosystem . This is not simply a list of unrelated commercial buyers. Corporate affiliations materially shape the observed procurement network. Historical UVZ disclosures explicitly identify Elektromashina as a major group subsidiary and document multiple other customers as UVZ-controlled enterprises. Second, the company demonstrates a technically heterogeneous product footprint . The same customer network contains specialized electromechanical equipment for armored vehicles, industrial lighting and control systems, and traction motors for civilian trams. This suggests a manufacturer whose public commercial footprint spans multiple applications rather than a single narrow component category. Third, customer intelligence cannot be reconstructed from procurement data alone . Court records and technical/product documentation materially improve the quality of attribution. The Kurganmashzavod cases demonstrate that a relationship can be followed from manufacturer, through direct customer, to specific product and then to downstream end user. The most defensible overall assessment is therefore: NPO Elektromashina's publicly observable customer network is broad in organizational scope but concentrated within a state-linked industrial ecosystem, with verified relationships covering both defense-related equipment and civilian transport/industrial applications. Public procurement data provide the backbone of the network, while court records, technical documentation and manufacturer/customer disclosures are necessary to resolve product identity, application and downstream relationships. The investigation does not establish the complete real-world customer base of NPO Elektromashina. It establishes the portion of that network that can be reconstructed from publicly accessible evidence with a defensible degree of confidence. That distinction is material: absence of a public procurement/product record should be treated as an evidence gap, not as evidence that the commercial relationship did not exist.

Sources

1. NPO Elektromashina corporate disclosure / issuer information

- legal identity, INN, OGRN, registered address. 2. Uralvagonzavod consolidated reporting and disclosures
- historical ownership and group structure, including NPO Elektromashina and major subsidiaries. 3. Synapse customer/procurement database
- 35 identified customers, 208 supplier-side contracts and aggregate contract values; used primarily for customer-universe construction. 4. EIS-derived procurement records via STAR
- individual contract numbers, dates, products, counterparties and values. 5. Arbitration Court / Garant - Kurganmashzavod vs. Elektromashina
- direct supply contract, actual deliveries and BU088-1S / 6EC088-1S. 6. Arbitration Court / Garant - 960-M DMB-4-M case

- downstream end user Military Unit 41450 and regulator failure. 7. Rostec - TAED72 / PK Transport Systems
- 2021 development and supply, testing, later deliveries and 2026 cooperation agreement. 8. AMASIS / Elektromashina - Omsktransmash project
- industrial lighting system, implementation work and customer testimonial. 9. Soyuzmash / industry publication - Uraltransmash 71-415R
- EDBT1 motor manufactured by Elektromashina and installed in the tram. 10. Google Patents - R15M-3S
- independent technical confirmation of Elektromashina manufacture and regulator function. Source-use note The procurement aggregators were used primarily for discovery and aggregation . Where an individual transaction or relationship was important to the analytical conclusion, the report prioritizes EIS-derived contract data, judicial records, corporate disclosures and

independent technical/industry sources . No relationship has been promoted to a higher confidence level solely because it appears in an aggregator.